

Module 4 Review

Permutations:

If we have *n options* and we want to *pick k* out of those, in a *certain order*, we can get the total number of ways of doing so by using the *Permutation formula*:

$$P(n, k) = \frac{n!}{(n - k)!}$$

Combinations:

In this case, the *order doesn't matter*:

$$C(n, k) = \frac{n!}{(n - k)! (k)!}$$

Probability:

$$\text{Probability of an event happening} = \frac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$$

Independent events:

$$\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$$

Binomial Distribution

- *p* : Probability of success
- *q* : Probability of failure; *q* = 1 - *p*
- *Bernoulli trial* : each repetition of an experiment involving only two outcomes
- *n* : Number of Bernoulli trials
- *r* : Number of successes

$$\Pr(X = r; n, p) = C(n, r)p^r(1 - p)^{(n-r)} = \frac{n!}{r!(n - r)!}p^r(1 - p)^{(n-r)}$$

Conditional Probability

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}} = \frac{P(A \cap B)}{P(B)}$$

Baye's rule

$$P(\text{Hypothesis}|\text{Evidence}) = \frac{P(\text{Evidence}|\text{Hypothesis}) \times P(\text{Hypothesis})}{P(\text{Evidence})}$$

Expectation

$$E[X] = \mu = \sum_{X=x} xP(X = x)$$

Random Variable

- **Discrete Random Variable:** A random variable that takes on a finite or countably infinite number of values. E.g. Results of a fair die roll
- **Continuous Random Variable:** A random variable that takes on a noncountable infinite number of values. E.g. Weight of a randomly chosen person

Probability Mass Functions (PMF): The probability distribution for a *discrete random variable* is usually called a PMF.

Properties of PMF:

- $f(x) > 0$ for x in sample space and 0 otherwise.
- $\sum_x f(x) = 1$. In other words, the sum of the probabilities of all possible outcomes of a given experiment equals to 1.

Probability Density Functions (PDF): the probability distribution function of a *continuous random variable* X is called a PDF.

Properties of a PDF:

- $f(x) > 0$ for x in sample space and 0 otherwise.
- The *area under the curve* is equal to 1.

Cumulative Distribution Function (CDF): A cumulative distribution function (CDF), usually denoted $F(x)$, is a function that gives the probability that the random variable, X , is less than or equal to the value x .

$$F(x) = P(X \leq x)$$

Variance of a Random Variable

$$\sigma^2 = Var(X) = \sum (x_i - \mu)^2 f(x_i)$$

*****We take each value of x , subtract the expected value, square that value and multiply that value by its probability. Then sum all of those values*****

The **Standard Deviation (σ)** of a random variable is the *square root of Variance*.

Properties of Variance and Expectation

▪ $E(c) = c$	▪ $Var(c) = 0$
▪ $E(cX) = c E(X)$	▪ $Var(cX) = c^2 Var(X)$
▪ $E(X + c) = E(X) + c$	▪ $Var(X + c) = Var(X)$
▪ $E(X + Y) = E(X) + E(Y)$	▪ $Var(X + Y) = Var(X) + Var(Y)$

	<i>Note: X and Y need to be independent</i>
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CLT for Sample Means

$\mu_x = \text{Mean of } X$

$\sigma_x = \text{Standard Deviation of } X$

$\bar{X} = \text{Oserved sample mean}$

$n = \text{sample size}$

The random variable $\bar{X}$ has a *z-score* associated with it which is given by:

$$z = \frac{\bar{x} - \mu_x}{\left(\frac{\sigma_x}{\sqrt{n}}\right)}$$

Here,

$\mu_x = \text{mean of } X \text{ and } \bar{X}$

$\frac{\sigma_x}{\sqrt{n}} = \sigma_{\bar{x}} = \text{Standard deviation of } \bar{X} \text{ or standard error of the mean}$

The *z-score* tells you *how many standard deviations away from the mean* a given observation is while a *z-table* shows the percentage or *probability of a value* being *less than a given z-score* in a standard normal distribution.

Confidence Interval

Confidence interval = sample mean $\pm$ margin of error

$$\text{Margin of Error} = z \times \frac{\text{population standard deviation}}{\sqrt{n}} = z \times \frac{\sigma}{\sqrt{n}}$$

A t-interval for the mean

When σ is unknown or when sample size is less than 30.

The unbiased estimate of the population standard deviation is represented using S .)

$$S = \sqrt{\sum_{i=1}^n \frac{1}{(n-1)} (X_i - \bar{X})^2}$$

d.o.f = $n-1$
 $n \rightarrow$ sample size

Here, $(n-1)$ is called the “*Degrees of freedom*”.

$$\text{Margin of Error} = t \times \frac{\text{unbiased estimate of } \sigma}{\sqrt{n}} = t \times \frac{S}{\sqrt{n}}$$

Let's Practice

1. A password must be a **4-digit code** (digits 0–9). Digits **can repeat**. How many passwords contain **at least one repeated digit**?
2. A password consists of **3 letters** followed by **2 digits**. Letters must be **distinct**; digits may repeat. How many possible passwords?
3. You have **4 math books**, **3 science books**, and **2 history books**. Books of the **same subject must stay together**. How many ways can all 9 books be arranged?
4. In a class of 40 students, 24 students take math, 18 take physics, and 10 take both. If a student is chosen at random, what is the probability the student takes **neither** subject?
5. A student guesses on a **5-choice** question. If the student's answer is **correct**, what is the probability they guessed rather than knew the answer? (Assume the student knows the answer with probability 0.7, and guesses otherwise.)
6. You flip a fair coin 9 times. What is the probability you get **more heads than tails**?

7. A password generator picks a 6-character password using uppercase letters (A–Z).

Let X = number of vowels in the password. Find $E[X]$.

8. A fair die is rolled 100 times. Let $\bar{X}$ be the sample mean.

Find $P(3.2 < \bar{X} < 3.6)$.

9. You sample 20 exam scores and get $\bar{X} = 76, s = 10$.

Construct a **99% CI** for the class mean.

10. You want to estimate average commute time with margin of error ± 3 **minutes** at **95% confidence**, and σ is known to be 12 minutes. What sample size is required?

Answers:

1. Total # of ways to generate the password with repeated digits :

$$10 \times 10 \times 10 \times 10 = 10^4 = 10,000$$

Total # of ways to generate the password without repeated digits :

$$10 \times 9 \times 8 \times 7 = 5040$$

$\therefore$ # of passwords with atleast 1 repeated digit :

$$10,000 - 5040 = 4960$$

2. Letters : we can choose 3 distinct letters :

$$26 \times 25 \times 24$$

Digits : 2 (repetitions allowed) digits :

$$10 \times 10$$

$$\therefore \text{Total} = 26 \times 25 \times 24 \times 10 \times 10 = 1560000$$

3. # of ways to arrange 4 maths books

$$4!$$

of ways to arrange 3 science books:

$$3!$$

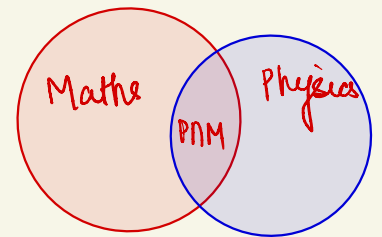
of ways to arrange 2 history books:

$$2!$$

of ways we can arrange Math, Science and History genre:

$$3!$$

$$\therefore \text{Total} : 4! \times 3! \times 2! \times 3! = 1728$$



4. Let M → student takes Maths
P → student takes Physics
 $P \cap M$ → student takes both

$$P(M) = \frac{24}{40}, \quad P(P) = \frac{18}{40}, \quad P(P \cap M) = \frac{10}{40}$$

$$\therefore P(\text{student takes neither}) = 1 - \left(\frac{24}{40} + \frac{18}{40} - \frac{10}{40} \right)$$
$$1 - \frac{32}{40} = \frac{8}{40} = \frac{1}{5}$$

5. $P(\text{knows}) = 0.7$

$$P(\text{guesses}) = 0.3$$

To find $P(\text{guesses} | \text{Correct})$

$$P(\text{guesses} | \text{Correct}) = \frac{P(\text{Correct} | \text{guesses}) \cdot P(\text{guesses})}{P(\text{Correct})}$$

$$P(\text{Correct} | \text{guesses}) = \frac{1}{5} \quad (\text{1 out of 5 questions})$$

$$P(\text{Correct} | \text{knows}) = 1 \quad (\text{if knows, will be correct})$$

$$P(\text{Correct}) = P(\text{Correct} | \text{knows}) \cdot P(\text{knows}) + P(\text{Correct} | \text{guesses}) \cdot P(\text{guesses})$$

$$= 1 \times 0.7 + \frac{1}{5} \times 0.3 = 0.76$$

$$\therefore P(\text{guesses} | \text{Correct}) = \frac{\frac{1}{5} \times 0.3}{0.76} = \frac{0.06}{0.76} = \frac{6}{76} = \frac{3}{38} \approx 0.079$$

6. Let X be the random variable for the # of heads in 9 coin flips.

$$\text{To find } P(X \geq 5) = P(X=5) + P(X=6) + P(X=7) + P(X=8) + P(X=9)$$

Using binomial:

$${}^9C_5 \times (0.5)^5 \times (0.5)^4 + {}^9C_6 \times (0.5)^6 \times (0.5)^3 + {}^9C_7 \times (0.5)^7 \times (0.5)^2 \\ + {}^9C_8 \times (0.5)^8 \times (0.5)^1 + {}^9C_9 \times (0.5)^9$$

$$= (0.5)^9 ({}^9C_5 + {}^9C_6 + {}^9C_7 + {}^9C_8 + {}^9C_9)$$

$$= \frac{1}{512} \times 256 = \frac{256}{512} = \frac{1}{2}$$

7. $P(\text{vowel}) = \frac{5}{26}$

Let X be the # of vowels in the password
for each position (1-6) let X_i be

$$X_i = \begin{cases} 1 & \text{if a vowel} \\ 0 & \text{if not a vowel} \end{cases}$$

Then

$$X = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$$

$$E[X_i] = 1 \times \frac{5}{26} + 0 \times \frac{21}{26} = \frac{5}{26}$$

$$\therefore E[X] = \frac{5}{26} + \frac{5}{26} + \frac{5}{26} + \frac{5}{26} + \frac{5}{26} + \frac{5}{26}$$

$$= 6 \cdot \frac{5}{26} = \frac{30}{26} = \frac{15}{13}$$

8. We can use CLT to solve this

$$E[X_i] = \mu = \left(\frac{1+2+3+\dots+6}{6} \right) = 3.5$$

(here X_i is the random variable associated with each coin toss)

$$\sigma^2 = 2.916 \text{ (using variance formula)}$$

$$\sigma = \sqrt{2.916} \approx 1.71$$

$$n = 100 \quad \therefore \sqrt{n} = 10$$

$$\therefore Z_{3.2} = \frac{3.2 - 3.5}{\frac{1.71}{10}} = -1.754$$

$$Z_{3.6} = \frac{3.6 - 3.5}{\frac{1.71}{10}} = 0.585$$

$$P(Z < 0.585) \approx 0.720$$

$$P(Z < -1.754) \approx 0.040$$

$$\therefore P(-1.754 < Z < 0.585) = 0.720 - 0.040 = 0.680$$

9. Given,

$$n = 20$$

$$S = 10$$

$$\bar{X} = 76$$

$$\text{Confidence} = 99\%$$

Since we are using S and $n < 30$, we will use t -distribution.

$$\text{Degrees of freedom} = 20 - 1 = 19$$

$$\text{Standard error} = \frac{S}{\sqrt{n}} = \frac{10}{\sqrt{20}} \approx 2.236$$

$$\text{Margin of error} = t \times 2.236 = 2.861 \times 2.236 \approx 6.40$$

$$\begin{aligned} \therefore \text{Confidence Interval} &= \bar{X} \pm 6.40 = 76 \pm 6.40 \\ &= (69.6, 82.4) \end{aligned}$$

10. Given ,

Margin of Error = 3

$$\sigma = 12$$

Confidence level = 95% $\therefore Z = 1.96$

$$\therefore Z \cdot \frac{\sigma}{\sqrt{n}} = 3$$

$$\Rightarrow \left(\frac{1.96 \times 12}{3} \right)^2 = n$$

$$\Rightarrow n = (7.84)^2 \approx 61.47$$

$\therefore n = 62$ (rounding off to the next whole number)